• By Theorem 2, each of the $n$ columns of $A$ must be pivot columns.
∴ By Theorem 2, the system of equations corresponding to the augmented matrix $[A\ b]$ for any $b$ must have a unique solution.
Similar to the above reasoning, the equation $T(x)=b$ has a unique solution for every $b$ in $\mathbb{R}^m$. By defn, $T$ is one-to-one. ∎

Book's proof

**Remark:** To prove a theorem that says "statement P is true if and only if statement Q is true," one must establish two things: (1) If P is true, then Q is true and (2) If Q is true, then P is true. The second requirement can also be established by showing (2a): If P is false, then Q is false. (This is called contrapositive reasoning.) This proof uses (1) and (2a) to show that P and Q are either both true or both false.

**Proof:** Since $T$ is linear, $T(0)=0$. If $T$ is one-to-one, then the equation $T(x)=0$ has atmost one solution and hence only the trivial solution. If $T$ is not one-to-one, then there is a $b$ that is the image of atleast two different vectors in $\mathbb{R}^n$ — say, $u$ and $v$. That is $T(u)=b$ and $T(v)=b$.
But then, since $T$ is linear,

$$\boxed{T(u-v)=T(u)-T(v)=b-b=0}$$

The vector $u-v$ is not zero, since $u \neq v$. Hence the equation $T(x)=0$ has more than one solution. So, either the two conditions in the theorem are both true or they are both false. ∎. (Very elegant proof)

**Theorem 12:** Let $T: \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation, and let $A$ be the standard matrix for $T$. Then:
a) $T$ maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of $A$ span $\mathbb{R}^m$;
b) $T$ is one-to-one if and only if the columns of $A$ are linearly independent.

**Proof:** a) By defn. of onto, for each vector $b$ in $\mathbb{R}^m$, the linear transformation $T(x)=b$ has atleast one solution.